

CIA TRIAD ANALYSIS

CYBER SECURITY TRACK — TASK 2

Name: Yuktha

Task: CIA Triad Analysis

Topic: Confidentiality, Integrity and Availability

1. INTRODUCTION

The CIA Triad is a fundamental model of information security. CIA stands for Confidentiality, Integrity and Availability. These three principles help protect sensitive information and ensure that computer systems and networks remain secure. Confidentiality prevents unauthorized access to information, integrity protects information from unauthorized modification, and availability ensures that information and services are accessible when needed. Together, these three principles form the foundation of an effective cybersecurity system.

2. THE THREE PRINCIPLES OF THE CIA TRIAD

2.1 Confidentiality

Confidentiality is the principle of protecting sensitive information from unauthorized access or disclosure. It ensures that only authorized users can access private data.

Example: A bank uses passwords, encryption and multi-factor authentication to protect customers' account details from unauthorized access.

Importance:

- Protects sensitive and personal information.
- Prevents unauthorized disclosure of data.
- Maintains user privacy and trust.
- Reduces the risk of identity theft.

2.2 Integrity

Integrity is the principle of maintaining the accuracy, consistency and reliability of information. It ensures that data is not changed, deleted or modified without proper authorization.

Example: A school uses access controls and audit logs to prevent unauthorized changes to students' examination marks.

Importance:

- Maintains the accuracy of information.
- Prevents unauthorized modification of data.
- Ensures that information remains trustworthy.
- Helps organizations make correct decisions.

2.3 Availability

Availability is the principle of ensuring that information, applications and systems are accessible to authorized users whenever required.

Example: A hospital maintains backup servers and backup power supplies to ensure that its patient records remain accessible during technical failures.

Importance:

- Ensures uninterrupted access to information.
- Reduces service downtime.
- Supports business and organizational operations.
- Helps maintain reliable services during emergencies.

3. FIVE REAL-WORLD EXAMPLES OF CYBERSECURITY INCIDENTS

3.1 Yahoo Data Breach (2013)

CIA principle: Confidentiality

Description: In 2013, Yahoo experienced a major data breach in which attackers stole user account information. The incident affected approximately 3 billion accounts, as Yahoo disclosed in 2017. The stolen information included names, email addresses and hashed passwords.

CIA principle affected: Confidentiality was compromised because unauthorized attackers gained access to users' private information.

Prevention:

- Use strong passwords and multi-factor authentication.
- Encrypt sensitive user information.
- Regularly update security systems.
- Monitor networks for suspicious activity.

3.2 Equifax Data Breach (2017)

CIA principle: Confidentiality

Description: In 2017, Equifax, a major credit reporting company, suffered a data breach that exposed the personal information of approximately 147 million people. The attackers exploited a vulnerability in a web application and accessed sensitive data, including names, birth dates and Social Security numbers.

CIA principle affected: Confidentiality was compromised because sensitive personal information was accessed by unauthorized individuals.

Prevention:

- Regularly update software and apply security patches.
- Perform vulnerability assessments.
- Use firewalls and intrusion detection systems.
- Encrypt sensitive personal information.

3.3 Stuxnet Attack (Discovered in 2010)

CIA principle: Integrity

Description: Stuxnet was a sophisticated computer worm discovered in 2010. It targeted industrial control systems and manipulated the operation of certain industrial equipment. It is widely associated with damage to Iranian nuclear centrifuges.

CIA principle affected: Integrity was compromised because the malware secretly changed the instructions and operation of industrial equipment.

Prevention:

- Secure industrial control systems.
- Restrict unauthorized access to critical systems.
- Monitor industrial equipment for unusual activity.
- Regularly audit system configurations.

3.4 WannaCry Ransomware Attack (2017)

CIA principle: Availability

Description: In May 2017, the WannaCry ransomware attack affected computers worldwide. It encrypted files and demanded ransom payments to restore access. The attack disrupted several organizations, including parts of the UK's National Health Service (NHS), affecting healthcare services.

CIA principle affected: Availability was compromised because users and organizations lost access to important files and computer systems. Integrity was also potentially affected.

Prevention:

- Regularly update operating systems.

- Maintain secure backups of important files.
- Use reliable antivirus software.
- Avoid opening suspicious email attachments.

3.5 Dyn DDoS Attack (2016)

CIA principle: Availability

Description: On 21 October 2016, Dyn, a major Domain Name System (DNS) service provider, experienced a large distributed denial-of-service (DDoS) attack. The attack overwhelmed its systems with traffic, disrupting access to many popular websites and online services.

CIA principle affected: Availability was compromised because users could not reliably access websites and services that depended on Dyn's DNS infrastructure.

Prevention:

- Use DDoS protection services.
- Deploy redundant DNS infrastructure.
- Monitor network traffic for unusual activity.
- Develop incident response and disaster recovery plans.

4. SUMMARY OF THE FIVE INCIDENTS

Cybersecurity incident	Year	CIA principle affected
Yahoo data breach	2013	Confidentiality
Equifax data breach	2017	Confidentiality
Stuxnet attack	2010	Integrity
WannaCry ransomware	2017	Availability
Dyn DDoS attack	2016	Availability

5. CONCLUSION

The CIA Triad is an essential part of cybersecurity because it helps organizations protect their information and computer systems. Confidentiality ensures that sensitive information is accessible only to authorized users. Integrity maintains the accuracy and reliability of information, while availability ensures that systems and services remain accessible when needed.

Real-world incidents such as the Yahoo and Equifax data breaches, the Stuxnet attack, WannaCry ransomware and the Dyn DDoS attack demonstrate how these principles can be

compromised. Organizations can reduce cybersecurity risks by implementing strong authentication, encryption, regular security updates, data backups and continuous monitoring.

Therefore, maintaining confidentiality, integrity and availability is essential for protecting information, preventing cyberattacks and ensuring the security and reliability of digital systems.

6. REFERENCES

1. Yahoo Security — Information about the Yahoo data breach.
2. Equifax — 2017 cybersecurity incident and data breach information.
3. CISA — Information about Stuxnet and industrial control system security.
4. NHS England — Information about the WannaCry ransomware incident.
5. Dyn — Information about the October 2016 DDoS attack.